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Epidemiology of bovine colostrum production in New York Holstein herds: Prepartum nutrition and metabolic indicators

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ABSTRACT

Colostrum yield and quality are influenced by prepartum nutrition and the metabolic status of the cow; however, data considering these associations on multiple dairy farms are limited. Our objective was to identify cow-level prepartum metabolic indicators, as well as farm-level nutritional strategies associated with colostrum yield and the indicator of colostrum quality, Brix %. A convenience sample of 19 New York Holstein dairies (median: 1,325 cows; range: 620 to 4,600 cows) were enrolled in this observational study. Records for individual colostrum yield and Brix % were collected by farm personnel between October 2019 and February 2021. Farms were visited 4 times, approximately 3 mo apart, to obtain feed samples of the prepartum diets, collect blood samples from 24 pre- and postpartum cows, respectively, and determine prepartum body condition score. Feed samples were submitted for analysis of chemical composition, and particle size was determined on-farm using a particle separator. Prepartum serum samples (n = 762) were analyzed for glucose and nonesterified fatty acid concentrations. Whole blood from postpartum cows was analyzed for herd prevalence of hyperketonemia (proportion of samples with β-hydroxybutyrate ≥1.2 mmol/L). A cohort of primiparous (PP; n = 1,337) and multiparous (MPS; n = 3,059) cows calving ± 14 d of each farm visit were included in the statistical analysis. Animals calving in this period were assigned results for the close-up diet composition and herd prevalence of hyperketonemia collected from the respective farm visit. Greatest colostrum yield from PP and MPS cows was associated with moderate starch [18.6–22.5% of dry matter (DM)] and a moderate herd prevalence of hyperketonemia (10.1–15.0%). Greatest colostrum yield from MPS cows was associated with moderate crude protein (13.6–15.5% of DM) and a less severe negative dietary cation-anion difference (DCAD; >−8 mEq/100 g), whereas greatest colostrum yield from PP cows was associated with low crude protein (≤13.5% of DM). In addition, a moderate proportion of the diet with particle length ≥19 mm (15.3–19.1%) was associated with lowest colostrum yield from PP and MPS cows. Highest colostrum Brix % was associated with prepartum dietary factors of low neutral detergent fiber (≤39.0% of DM) and high proportion of the diet with particle length ≥19 mm (>19.1%). In addition, low starch (≤18.5% of DM) and low and moderate DCAD level (≥−15.9 mEq/100 g) were associated with greatest Brix % from PP cows, whereas moderate DCAD (−15.9 to −8.0 mEq/100 g) was associated with greatest Brix % from MPS cows. Prepartum serum nonesterified fatty acid concentration ≥290 μEq/L was associated with increased colostrum yield, but prepartum serum glucose concentration and body condition score were not associated with colostrum yield or Brix %. These data provide nutritional and metabolic variables to consider when troubleshooting colostrum production on farms.

Key words: colostrum, Brix %, nutrition, metabolism

INTRODUCTION

Mammary colostrogenesis is a metabolic process with intensive nutrient requirements that occurs during the last weeks of gestation (McGuirk and Collins, 2004). Advancements in dry cow management and prepartum nutritional strategies have improved postpartum health and milk production (Van Saun and Sniffen, 2014; Cardoso et al., 2020), yet low colostrum yields have been reported, particularly in the fall and winter months (Gavin et al., 2018; Borchardt et al., 2022, Westhoff et al., 2023). We lack knowledge on the effect of different dry cow feeding strategies and the metabolic status of cows on colostrum yield, as well as Brix % as an indicator of IgG concentration. Feeding a controlled energy or low starch prepartum diet has been reported to increase colostrum IgG concentration (Mann et al., 2016; Fischer-Tlustos et al., 2021). Altering prepartum protein supply did not affect colostrum yield (Amira-

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badi Farahani et al., 2017, 2019), but the effect on IgG concentration has not been investigated with current prepartum protein feeding strategies. Authors of previous studies report that manipulation of DCAD did not influence colostrum IgG concentration; however, results on colostrum yield were mixed (Weich et al., 2013; Lopera et al., 2018; Martinez et al., 2018). In a recent study, high-quality colostrum (IgG ≥50 g/L) was associated with greater prepartum glucose concentrations (Rossi et al., 2023), but relatively little is known about associations of prepartum metabolic status and colostrum production in dairy cattle.

Given the lack of observational and experimental data considering prepartum nutrition and indicators of metabolic status, our ability to understand variables to consider during periods of low colostrum yield or quality is limited. We hypothesized that colostrum yield and Brix % are associated with the herd's nutritional strategy as well as cow-level prepartum indicators of metabolic status. Our objective of this work performed on NY Holstein dairy farms was to identify prepartum metabolites and diet components associated with colostrum yield and Brix %.

MATERIALS AND METHODS

Farm Characteristics

All procedures were approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2019–0058). Study design describing details of farm enrollment, a detailed management survey, colostrum records, and farm visits was as previously described (Westhoff et al., 2023). Briefly, 19 New York Holstein herds were enrolled in this observational study between October 2019 and February 2021 as a convenience sample. Herd inclusion criteria were (1) ability of farm personnel to collect and record individual colostrum weight or volume and a composite sample digital Brix % reading, (2) minimum herd size of 500 lactating Holstein cows, (3) use of dairy management software DairyComp 305 (DC305, Valley Ag Software), and (4) heifer calves housed on site for at least the first week of life. Colostrum was harvested by farm personnel according to the existing farm protocol, and colostrum yield and Brix % were recorded. Farms were visited 4 times, approximately 3 mo apart, during the data collection period.

Prepartum Diet Samples and Analysis

Diets fed to heifers from −60 to 0 d relative to parturition and those fed to dry cows were sampled at each visit. Particle size was determined by one investigator with a 4-sieve Penn State Particle Separator (**PSPS**, Nasco). Samples of the diets were submitted to a commercial laboratory (Dairy One Cooperative Inc.) for prediction of chemical composition by near-infrared reflectance spectroscopy (FOSS NIRSystems with calibrations according to principles for AOAC methods; Dairy One, 2022) and wet chemistry analysis of minerals (procedures based on CEM Application Notes for sample digestion followed by inductively coupled plasma radial spectrometry; Dairy One, 2022). Physically effective NDF (**peNDF**) was calculated by multiplying diet amylase (**a**)NDF (% of DM) by the proportion of the diet ≥4 mm (Kmicikewycz et al., 2015).

Prepartum Blood Samples and Analysis

Blood samples were collected from a convenience sample of 8 primiparous (**PP**) and 16 multiparous (**MPS**) cows in the prepartum period (3–14 d before expected parturition). Blood was collected from the coccygeal vein or artery into an evacuated collection tube without additive (Becton Dickinson) and allowed to clot at ambient temperature for 30 min before being placed on ice for transport to the laboratory. Serum was harvested within 12 h by centrifugation at 2,300 × *g* at 4°C for 15 min and stored at −20°C until analysis. Serum concentrations of glucose and nonesterified fatty acids (**NEFA**) were determined in duplicate by enzymatic colorimetric analysis [PGO enzyme preparation, Millipore Sigma; HR Series NEFA-HR (2), Wako Life Sciences] for all samples from prepartum cows with a recorded colostrum yield and Brix %. In-house samples were run on every plate for quality control. The intra- and interassay coefficient of variation determined from 762 samples were 2.4 and 4.8% for glucose and 4.4 and 9.4% for NEFA, respectively.

Prepartum BCS

Body condition score was determined on the same prepartum animals as described above by a single trained investigator using a 5-point scale and 0.25-point increments (Edmonson et al., 1989).

Herd Prevalence of Postpartum Hyperketonemia

As an indicator of overall success of metabolic adaptation during the transition period, the herd prevalence of postpartum hyperketonemia (BHB ≥1.2 mmol/L; McArt et al., 2012) was determined at each visit from a convenience sample of 8 PP and 16 MPS postpartum cows (3–14 DIM). Sodium heparinized whole blood was obtained from the coccygeal vessels as described above. Samples were immediately placed on ice for

transport to the laboratory, and warmed on arrival to 37°C before using a Nova Vet handheld ketone meter (Nova Biomedical) with a 1.25 calibration slope factor as previously described (Bach et al., 2016). Calibration of the Nova Vet ketone meter was tested before every use with Nova Max Ketone/Glucose Control Solution (Nova Biomedical).

Analytical Approach

Sample size of 24 pre- and postpartum animals for blood sampling was as previously described (Kerwin et al., 2022). For statistical analysis, we used models to explore associations of herd-level and individual animal associations with colostrum production, respectively.

Close-up diet composition and postpartum hyperketonemia prevalence were considered for associations with colostrum yield and Brix % at the herd level. The top screen of the PSPS (particle length ≥19 mm) was used as an indicator of forage chop length in the prepartum diet. Additionally, the herd's prevalence of hyperketonemia at each visit was used as an indicator of overall success of metabolic adaptation that could influence colostrum production. A cohort of animals calving ± 14 d of each farm visit were included in the statistical analysis. Animals calving in this period were assigned results for the close-up diet composition and particle length from the pen they were housed in, respectively, as well as the herd's prevalence of postpartum hyperketonemia from the respective farm visit. Variables were independently assessed for a linear relationship with colostrum yield and Brix %. If the assumption of a linear relationship was not fulfilled (defined as an absolute correlation coefficient ≥0.20), variables were categorized for subsequent analysis. Quartiles 1 and 3 were used as cut points to categorized diet composition for starch (≤18.5, 18.6–22.5, >22.5) % of DM, aNDF (≤39.0, 39.1–43.5, >43.5) % of DM, CP (≤13.5, 13.6–15.5, >15.5) % of DM, DCAD (≤−16.0, −15.9 to −8.0, >−8.0) mEq/100 g, and physically effective NDF (**peNDF**; ≤27.0, 27.1–32.0, >32.0). Proportion of the diet with particle length ≥19 mm was categorized by quartile (≤11.2, 11.3–15.2, 15.3–19.1, >19.1) % and prevalence of hyperketonemia was grouped into even categories (≤5.0, 5.1–10.0, 10.1–15.0, >15.0) % considering the herd-alarm level associated with negative health events (Ospina et al., 2010a).

Prepartum BCS and concentrations of glucose and NEFA were considered for associations with colostrum yield and Brix % at the cow level. Variables were categorized if the assumption of a linear relationship was not fulfilled as described above. Cows were categorized by serum concentrations of NEFA (<290, ≥290) μEq/L defined as elevated and associated with negative postpartum health and production (Ospina et al., 2010b) and glucose was categorized by quartile (≤58.0, 58.1–63.0, 63.1–68.0, >68.0 mg/dL). Due to fewer cows with BCS <3.00 and >3.75, BCS was categorized as ≤3.00, 3.25–3.50, and ≥3.75.

For statistical analysis of herd-level variables, mixed effects multivariable models with repeated measures were generated using PROC MIXED (SAS v. 9.4) for the outcome variables colostrum yield and Brix %. All models were controlled for close-up pen stocking density, whether the close-up pen commingled primiparous and multiparous cows, and month of farm visit as well as far-off diet content of starch, CP, aNDF, DCAD, peNDF, and proportion of the diet with particle length ≥19 mm. The random effect of parity, repeated effect of visit, subject of farm, and compound symmetry covariance structure were used for all models. Independent variables were assessed for multicollinearity through the variance inflation factor, tolerance, and eigensystem analysis of the correlation matrix in PROC REG (SAS v. 9.4; Schreiber-Gregory and Jackson, 2017). If variables had a tolerance <0.1, variance inflation factor >10, and an eigenvalue corresponding condition index >100, the variable with the largest variance inflation factor was removed. Close-up diet components and prevalence of hyperketonemia were screened in univariable models and entered the initial multivariable model when $P \leq 0.20$. Stepwise manual backward elimination was used until all remaining variables had $P \leq 0.05$. The interaction with parity group (1 vs. ≥2) was then tested. Interaction terms remained in the model when $P \leq 0.05$ and the SLICE option was used to determine at which parity group the interaction occurred. The assumptions of normality and homoscedasticity of the residuals were visually assessed, and colostrum yield was transformed using the natural logarithm to meet the model assumptions. Tukey's post hoc test was used to adjust for the number of pairwise comparisons.

For statistical analysis of cow-level variables, mixed effect multivariable models (PROC MIXED) including the fixed effects of BCS, and serum concentrations of glucose and NEFA were built for the dependent variables of colostrum yield and Brix %. Models included the random effect of herd and were controlled for parity group (1 vs. ≥2), month of farm visit, and days relative to calving. To fulfill the model assumptions of normality and homoscedasticity of the residuals, colostrum yield was transformed using the natural logarithm.

RESULTS

Enrolled herds, colostrum production, the study population, and farm visits were described previously (Westhoff et al., 2023). Briefly, 13 farms were visited

3 times and 6 farms were visited 4 times during the study due to a Cornell University mandated COVID-19 shutdown. Diets and blood samples were not collected at farm A and staff at farm T were unable to record colostrum yield and Brix %; therefore, final analysis was performed on data from 57 farm visits on 17 farms.

Prepartum Nutrition and Prevalence of Hyperketonemia at Each Visit

Two producers fed separate close-up diets to heifers and cows resulting in analysis of 64 close-up diets. Descriptive statistics for close-up diets and prevalence of hyperketonemia are presented in Table 1 and Table 2. Final analysis was performed on PP (n = 1,337) and MPS (n = 3,059) cows calving ± 14 d of the farm visit with >0-kg recorded colostrum yield. After univariable screening, the variables offered to the initial multivariable model for colostrum yield included close-up diet content of starch (categorized), aNDF (categorized), peNDF (categorized), crude protein (categorized), DCAD (categorized), proportion of the diet with particle length ≥19 mm (categorized), and prevalence of hyperketonemia (categorized). The final model is presented in Table 3. Colostrum yield was associated with diet content of starch ($P < 0.01$), CP ($P < 0.01$), DCAD ($P < 0.01$), proportion of the diet with particle length ≥19 mm ($P < 0.01$), and prevalence of hyperketonemia ($P < 0.01$). A moderate herd prevalence of hyperketonemia (10.1–15.0%) was associated with the greatest colostrum yield. Diet content × parity group (1 vs. ≥2) interactions were identified for starch ($P < 0.01$), CP ($P < 0.01$), DCAD ($P < 0.01$), and proportion of the diet with particle length ≥19 mm ($P < 0.01$; Figure 1) such that greatest colostrum yield from MPS cows was associated with moderate CP (13.6–15.5% of DM) and a less severe negative DCAD (>−8 mEq/100 g) whereas greatest colostrum yield from PP cows was associated with low CP (≤13.5% of DM) and colostrum yield from PP cows was not associated with DCAD ($P = 0.06$). In addition, lowest colostrum yield from MPS cows was associated with low starch (≤18.5% of DM), but lowest colostrum yield from PP cows was associated with high starch (>22.5% of DM). For colostrum Brix %, diet content of starch (categorized), aNDF (categorized), DCAD (categorized), proportion of the diet with particle length ≥19 mm (categorized), and prevalence of hyperketonemia (categorized) were offered to the initial multivariable mixed model after univariable screening. Variables remaining in the final multivariable model (Table 3) for colostrum Brix % were diet content of starch ($P < 0.01$), aNDF ($P < 0.01$), DCAD ($P < 0.01$), and proportion of the diet with particle length ≥19 mm ($P = 0.02$). Colostrum Brix % was greatest with low aNDF (≤39.0% of DM) and a high proportion of the diet with particle length ≥19 mm (>19.1%). Diet content × parity group interactions were identified for starch ($P < 0.01$) and DCAD ($P < 0.01$; Figure 2) such that greatest colostrum Brix % from PP cows was associated with low starch (≤18.5% of DM) and low or moderate DCAD (>−15.9 mEq/100 g), whereas greatest colostrum Brix % from MPS cows was associated with moderate DCAD (−15.9 to −8.0 mEq/100 g) and was not associated with starch ($P = 0.07$).

Prepartum Metabolites and BCS

Descriptive statistics for BCS, and serum concentrations of glucose and NEFA from prepartum cows (n = 762) are reported in Table 4. Colostrum yield was not associated with prepartum glucose concentration ($P = 0.69$) or BCS ($P = 0.95$), but greater yield was associated with prepartum NEFA ≥290 μEq/L compared with <290 μEq/L [LSM (95% CI); 4.7 (3.9–5.7) vs. 4.1 (3.5–4.9) kg; $P = 0.03$, respectively]. Colostrum Brix % was not associated with prepartum BCS ($P = 0.17$), glucose ($P = 0.68$), or NEFA ($P = 0.45$).

Table 1. Descriptive statistics for close-up diets (n = 64) and herd prevalence of hyperketonemia collected on 57 farm visits from a convenience sample of 17 New York dairy farms

Variable	Mean	Median	SD	Range
Starch (% of DM)	20.7	21.0	3.1	13.4 to 29.4
aNDF[1] (% of DM)	41.3	41.5	3.3	33.5 to 50.9
CP (% of DM)	14.6	14.3	1.6	11.5 to 18.5
DCAD (mEq/100 g)	−11.6	−12.0	7.3	−30 to 7
peNDF[2]	30.2	30.1	4.1	21.5 to 40.2
≥19 mm (% AF)[3]	15.7	15.4	5.7	6.5 to 32.9
Prevalence of hyperketonemia[4] (%)	10.4	8.3	9.4	0 to 45.8

[1]aNDF = amylase NDF.

[2]Physically effective NDF calculated by multiplying diet aNDF by percent of diet ≥4 mm.

[3]Percent of diet with particle length ≥19 mm. AF = as fed.

[4]Percent of fresh cows (3–14 DIM) with BHB ≥1.2 mmol/L.

Table 2. Descriptive statistics for categorized close-up diets (n = 64) and herd prevalence of hyperketonemia collected on 57 farm visits from a convenience sample of 17 New York dairy farms

Variable	n[1] (%)	Primiparous[2] (%)	Multiparous[2] (%)
Starch (% of DM)			
≤18.5	15 (23.4)	274 (20.5)	437 (14.3)
18.6–22.5	34 (53.2)	828 (61.9)	1,855 (60.6)
>22.5	15 (23.4)	235 (17.6)	767 (25.1)
aNDF[3] (% of DM)			
≤39.0	16 (25.0)	344 (25.7)	848 (27.7)
39.1–43.5	31 (48.4)	657 (49.2)	1,366 (44.7)
>43.5	17 (26.6)	336 (25.1)	845 (27.6)
CP (% of DM)			
≤13.5	16 (25.0)	395 (29.5)	1,121 (36.6)
13.6–15.5	30 (46.9)	694 (52.0)	1,351 (44.2)
>15.5	18 (28.1)	248 (18.5)	587 (19.2)
DCAD (mEq/100 g)			
≤−16	17 (26.5)	368 (27.5)	852 (27.9)
−15.9 to −8.0	32 (50.0)	662 (49.5)	1,685 (55.0)
>−8	15 (23.5)	307 (23.0)	522 (17.1)
peNDF[4]			
≤27.0	13 (20.3)	197 (14.7)	514 (16.9)
27.1–32.0	32 (50.0)	717 (53.7)	1,685 (55.1)
>32.0	19 (29.7)	423 (31.6)	860 (28.1)
≥19 mm (% AF)[5]			
≤11.2	15 (23.4)	337 (25.2)	640 (20.9)
11.3–15.2	16 (25.0)	301 (22.5)	796 (26.0)
15.3–19.1	16 (25.0)	456 (34.1)	746 (24.4)
>19.1	17 (26.6)	243 (18.2)	877 (28.7)
Prevalence of hyperketonemia[6] (%)			
≤5.0	24 (42.2)	661 (49.4)	1,254 (36.8)
5.1–10.0	8 (14.0)	244 (18.3)	465 (16.8)
10.1–15.0	9 (15.8)	196 (14.7)	723 (26.2)
>15.0	16 (28.0)	236 (17.6)	617 (20.2)

[1]Number of close-up diets or farm visits.

[2]Diet composition and prevalence of hyperketonemia were applied to animals calving ±14 d of the farm visit.

[3]aNDF = amylase NDF.

[4]Physically effective NDF calculated by multiplying diet aNDF by percent of diet ≥4 mm. AF = as fed.

[5]Percent of diet with particle length ≥19 mm.

[6]Percent of fresh cows (3–14 DIM) with BHB ≥1.2 mmol/L.

DISCUSSION

Our objective was to identify herd-level prepartum diet components and cow-level prepartum metabolites associated with colostrum yield and Brix %. In this study, prepartum diets with CP 13.6 to 15.5% of DM and ≤13.5% of DM were associated with the greatest colostrum yield from MPS and PP cows, respectively. In contrast, Amirabadi Farahani et al. (2017, 2019) reported no difference in colostrum yield from MPS cows when feeding prepartum diets with varying levels of CP. In agreement with the current study, several authors reported that colostrum quality was not affected by prepartum protein supply (Santos et al., 2001; Hare et al., 2019). With recent advancements in balancing diets for EAA (LaPierre et al., 2019), future studies are needed to understand the role of balanced AA supply on colostrum production.

Feeding a negative DCAD diet to prepartum cows is common practice due to reported improvements in hypocalcemia prevention, postpartum production, and overall reduced disease incidence, but reductions in DMI have been reported (Santos et al., 2019). In the current study, a lower DCAD level was associated with the lowest Brix % from PP and MPS cows. In contrast, authors of previous studies have reported colostrum IgG concentration was not affected by DCAD level (Lopera et al., 2018; Martinez et al., 2018). In agreement with the current study, colostrum yield from MPS cows was lower from cows fed diets with reduced DCAD (−7 vs. −18 mEq/100 g of DM; Lopera et al., 2018), although authors of other studies have reported no difference in colostrum yield when cows were fed +13 versus −13 and +12 versus −16 mEq/100 g of DM (Weich et al., 2013; Martinez et al., 2018). Although DMI was not available in the current study, Lopera et al. (2018) hypothesized reduced DMI due to a more severe metabolic acidosis could affect colostrogenesis.

Controlled energy, high forage prepartum diets have been widely adopted for the improvements in postpar-

Table 3. Mixed effects multivariable models for close-up diet content and herd prevalence of hyperketonemia associated with colostrum yield and Brix % from primiparous (n = 1,337) and multiparous (n = 3,059) Holstein cows from 17 New York farms

	Colostrum yield[1] (kg)		Colostrum Brix %[2]	
Variable	LSM (95% CI)[3]	*P*	LSM (95% CI)[3]	*P*
Starch (% of DM)		<0.01		<0.01
≤18.5	3.4 (2.7–4.1)[b]		26.4 (24.7–28.1)[a]	
18.6–22.5	4.5 (3.9–5.2)[a]		25.0 (23.4–26.6)[b]	
>22.5	3.7 (3.1–4.4)[b]		24.8 (23.1–26.5)[b]	
aNDF[4] (% of DM)				<0.01
≤39.0			24.4 (24.4–27.7)[a]	
39.1–43.5			23.9 (23.9–27.1)[ab]	
>43.5			23.0 (23.0–27.1)[b]	
CP (% of DM)		<0.01		
≤13.5	4.1 (3.4–4.9)[a]			
13.6–15.5	4.3 (3.6–5.1)[a]			
>15.5	3.1 (2.5–3.8)[b]			
DCAD (mEq/100 g)		<0.01		<0.01
≤−16	3.3 (2.7–4.1)[b]		24.6 (22.9–26.3)[b]	
−15.9 to −8.0	3.6 (3.0–4.3)[b]		25.7 (24.1–27.3)[a]	
>−8	4.6 (3.8–5.5)[a]		25.9 (24.3–27.6)[a]	
≥19 mm (% AF)[5]		<0.01		0.02
≤11.2	3.9 (3.2–4.8)[a]		25.1 (23.5–26.8)[ab]	
11.3–15.2	4.0 (3.3–4.8)[a]		25.0 (23.3–26.6)[b]	
15.3–19.1	3.1 (2.6–3.8)[b]		25.5 (23.8–27.2)[ab]	
>19.1	4.3 (3.5–5.2)[a]		26.0 (24.4–27.6)[a]	
Prevalence of hyperketonemia[6] (%)		<0.01		
≤5.0	3.7 (3.1–4.4)[b]			
5.1–10.0	3.6 (3.0–4.4)[b]			
10.1–15.0	4.5 (3.7–5.4)[a]			
>15.0	3.4 (2.7–4.3)[b]			
Starch × parity group[7]		<0.01		<0.01
CP × parity group[7]		<0.01		
DCAD × parity group[7]		<0.01		<0.01
≥19 mm × parity group[7]		<0.01		

[a,b]Least squares means (95% CI) with different superscripts differ ($P \leq 0.05$; Tukey's test).

[1]Data natural logarithm transformed before analysis. Reported as back-transformed LSM (95% CI).

[2]Colostrum yield was included as a covariate.

[3]Model included a repeated effect of farm visit, subject of farm, and was controlled for close-up pen stocking density, whether primiparous and multiparous animals were commingled in the close-up pen, parity group, month of farm visit, and far-off diet content of starch, aNDF, CP, DCAD, peNDF, and percent of the diet with particle length ≥19 mm.

[4]aNDF = amylase NDF.

[5]Percent of diet with particle length ≥19 mm. AF = as fed.

[6]Percent of fresh cows (3–14 DIM) with BHB ≥1.2 mmol/L.

[7]Parity group = 1 vs. ≥2.

tum metabolic health (Janovick et al., 2011; Mann et al., 2015). In the current study, starch content was associated with colostrum yield. Diets containing greater levels of ME have been shown to increase colostrum yield from sheep and cows (McGovern et al., 2015; Dunn et al., 2017). In addition, late gestation short-term supplementation of cereal grains was shown to increase colostrum yield from sheep and the authors hypothesized that greater energy intake could have increased the rate of progesterone clearance (Banchero et al., 2015). To date, the role of progesterone in bovine colostrogenesis remains inconclusive (Gross et al., 2014; Baumrucker et al., 2021). Colostrum IgG concentration was greatest in cows fed a controlled energy level during the prepartum period (Mann et al., 2016; Fischer-Tlustos et al., 2021). In contrast, dietary starch content was not associated with Brix % from MPS cows in the current study; however, greatest colostrum Brix % from PP cows was associated with low dietary starch content. Similarly, Dunn et al. (2017) did not observe a difference in colostrum IgG concentration from MPS cows but reported greater circulating serum IgG concentration in cows supplemented with concentrate to increase energy intake during the dry period. The mechanisms contributing to decreased colostrum Brix % from PP cows observed in the current study remains to be understood (Mann et al., 2016).

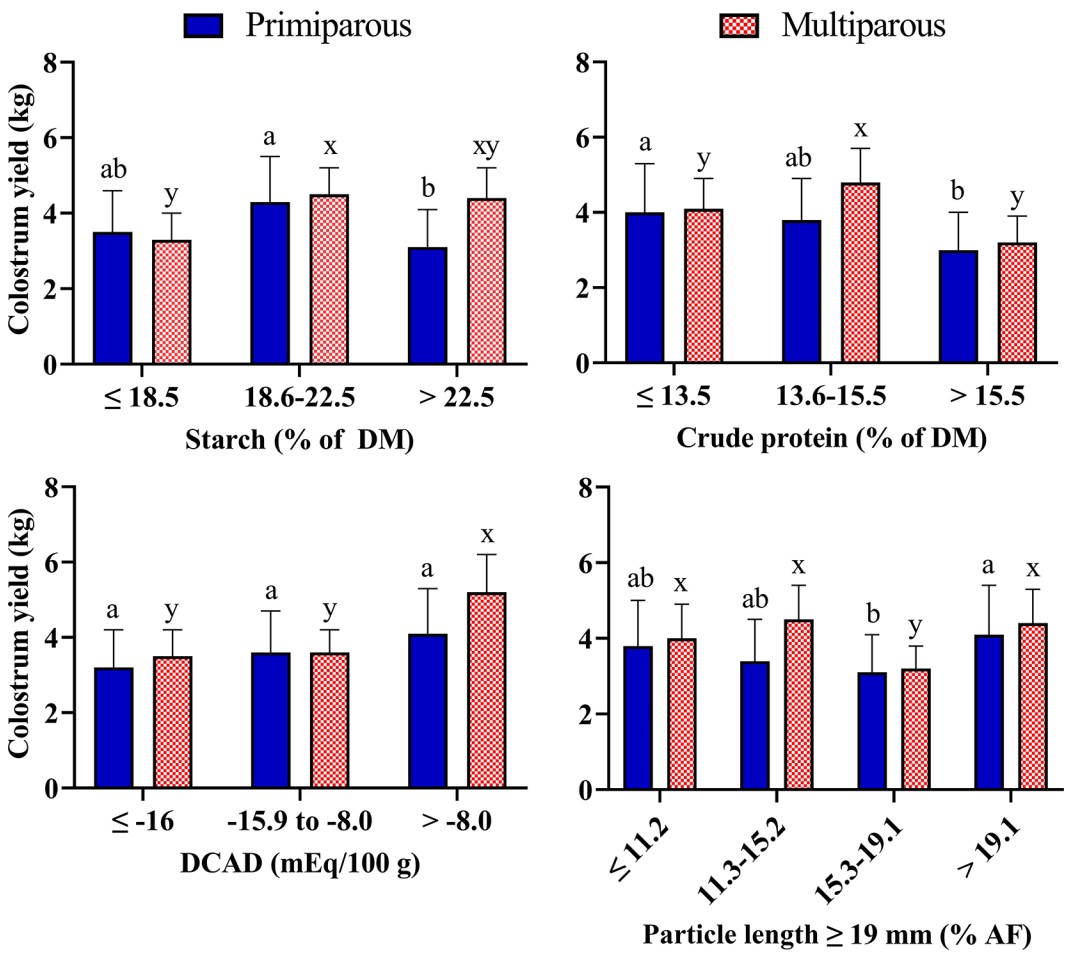


Figure 1. Close-up diet content of starch, CP, DCAD, and percent of the diet with particle length ≥19 mm associated with colostrum yield from primiparous (n = 1,337) and multiparous (n = 3,059) Holstein cows from 17 New York farms. Back-transformed LSM ± 95% CI with different letters differ ($P \leq 0.05$; Tukey's test). Starch × parity group (1 vs. ≥2), $P < 0.01$; CP × parity group, $P < 0.01$; DCAD × parity group, $P < 0.01$; particle length ≥19 mm × parity group, $P < 0.01$. AF = as fed.

Colostrum yield and Brix % were associated with the proportion of the diet in the top sieve of the PSPS (particle length ≥19 mm), used in this study as an indicator of long forage particles commonly observed in prepartum diets with greater inclusion of bulky forages. Physical sorting of the diet results in changes in feeding behavior (Greter and Devries, 2011). Although colostrum yield and quality were not reported, Havekes et al. (2020) showed reducing straw chop length in the prepartum diet increased DMI and reduced physical sorting, and the relative drop in DMI during the week before calving. We were unable to consider the change in DMI and the extent of physical sorting of the diets in the current study; therefore, their contribution to the observed associations would require further work.

Table 4. Descriptive statistics for prepartum BCS and indicators of metabolic status from a convenience sample of Holstein cows (n = 762)

Variable[1]	Mean	Median	SD	Range
Glucose (mg/dL)	63.4	63.1	9.5	37.2–101.0
NEFA[2] (μEq/L)	215.2	177	142.6	31.6–1,728.4
BCS[3]	3.32	3.25	0.34	2.00–5.00

[1]Determined in prepartum cows 3 to 14 d before expected calving.
[2]NEFA = nonesterified fatty acids.
[3]Determined using a 5-point scale with 0.25-point increments.

At the cow level, prepartum BCS and concentrations of glucose and NEFA were explored for associations with colostrum yield and Brix %. In agreement with the current study, previous observational studies have not found an association between BCS and colostrum yield or quality (Conneely et al., 2013; Denholm et al., 2018; Immler et al., 2021). Rossi et al., (2023) showed cows producing colostrum with IgG ≥50 g/L were associated with greater glucose concentrations compared with cows producing colostrum with IgG <50 g/L. The present study did not find an association between

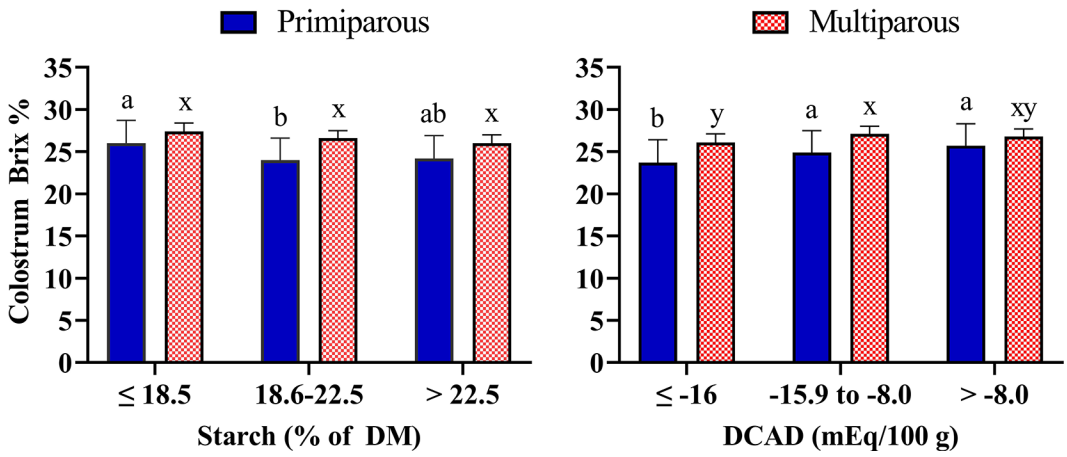


Figure 2. Close-up diet content of starch and DCAD associated with colostrum Brix % from primiparous (n = 1,337) and multiparous (n = 3,059) Holstein cows from 17 New York farms. LSM ± 95% CI with different letters differ ($P \leq 0.05$; Tukey's test). Starch × parity group (1 vs. ≥2), $P \leq 0.01$; DCAD × parity group, $P \leq 0.01$.

prepartum glucose concentration and colostrum Brix %. Sampling cows at a single time point 3 to 14 d before expected calving may have limited our ability to find such an association if present. Greater prepartum serum concentration of NEFA (≥290 μEq/L) was associated with increased colostrum yield in the current study. In contrast, Rossi et al., (2023) reported no difference in prepartum NEFA concentration from cows producing ≥6 or <6 L of colostrum, but cows producing ≥6 L were associated with an increased prepartum BHB concentration. Circulating ketone bodies, as an indicator of metabolic adaptation, have been associated with health events and milk production at the cow and herd level (Ospina et al., 2010a; McArt et al., 2013) and can be influenced by prepartum nutrition and management practices (Ospina et al., 2010a; Mann et al., 2015; Cardoso et al., 2020). Although prepartum BHB was not available in the present study, prevalence of hyperketonemia during early lactation (3–14 DIM) was associated with colostrum yield. Vanholder et al. (2015) reported greater colostrum yield was associated with an increased risk of subclinical and clinical ketosis. Colostrum formation is an energy intensive process requiring 1.1 Mcal of energy to produce 1 kg of colostrum (McGuirk and Collins, 2004) and the increased energy demand to produce more colostrum could contribute to the observed associations.

The farms and sampled animals in this study population were a convenience sample. Given our study design and frequency of visits, short-term changes in diets or prevalence of hyperketonemia were not accounted for. In the current study, data on individual urine pH as an indicator of successful acidification when using DCAD were not collected, and metabolizable protein content of the prepartum diets could not be estimated; therefore, analyzed DCAD and CP of the diets were used for statistical analysis. Complete information was not available to control for facility designs, days in the close-up pen, individual feed additives included in each prepartum diet, as well as frequency of feed delivery and feed push up which limits our ability to associate them with outcomes of interest in this study. In addition, the administration of oxytocin to some cows at colostrum harvest, reported as a management strategy by 11 (57.9%) producers (Westhoff et al., 2023) was not accounted for as a potential confounding variable in the current analysis as we could not ascertain that individual cows did or did not receive oxytocin. Nutrition and farm management were reflective of mid- to large-size New York dairy farms and caution should be taken when translating these results to farms with different management strategies.

CONCLUSIONS

Colostrum yield and Brix % were associated with prepartum NEFA concentration, diet components, and prevalence of hyperketonemia. Although other variables should be considered, these data provide prepartum variables to consider when troubleshooting colostrum production on farms.

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